AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 23 · STUDENT LESSON

Hold Out Test Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Freeze the version. Record the exposure. Bound the claim.

Interpret final-test evidence and report test-informed revisions honestly.

Keep one habit: Frozen20-case record → pictured version boundary → complete court actions → no-fresh-data report → independent new counts.

MAIN LESSON · SECTION 1

Which model version did the final test actually evaluate?

> A revised model needs a separate claim from the version tested before its errors were inspected.

Learning goals

I can **freeze** the full evaluation procedure before testing.

I can report a held-out result with counts and limitations.

I can recognize when test feedback has become development.

Recall development versus untouched evaluation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.
- Why is validation not an untouched final test?

End goal

By the end, I can explain hold out test data using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.



LEARNING TARGET After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Record the freeze before reading final outcomes

> A held-out set’s value depends on its actual use,not the folder name.

Freeze contract and fictional scope

A final test is valuable because it did not guide model choices. **Freeze** the model, **training** pre**process**ing, threshold, metric, missing-**prediction** policy and test membership. Record the version before opening test labels.

A classroom result is **evidence** about a small chosen set, not approval for deployment.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines final testing and honest **claim**s.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use hold out test data to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Calculate the supplied final-record metrics and bound their claim

> Separation protects the comparison from choice-driven reuse; it does not certify the dataset.

Complete20-case frozen-result example

A frozen classifier is tested on 20 fictional cases: TN 12 FP 2 FN 3 TP 3, blocked positive. Accuracy is 15/20=75 percent, precision 3/5=60 percent and recall 3/6=50 percent.

Its previously fitted always-clear baseline gets 14/20=70 percent. The candidate improves accuracy here but misses half the blocked cases.

State version and worked-simulation assumptions

The complete classroom record assumes version V1,its training-derived always-clear baseline,and all prediction/metric/missing-case rules were frozen before these outcomes were inspected. Because the labels are printed here,this is a worked simulation of a prior test record,not a genuinely hidden final test for a learner’s new model. A later threshold-adjusted V2cannot claim these same cases as untouched.

Preserve the original V1result and the full exposure history.

Explain without strengthening the counts

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger **claim**.

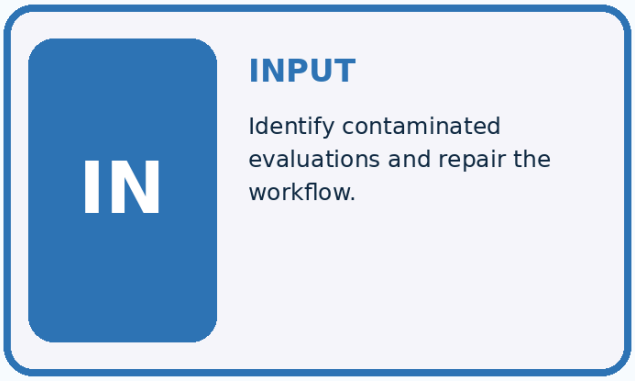
Keep the distinction between an observed result and a proposed next step.

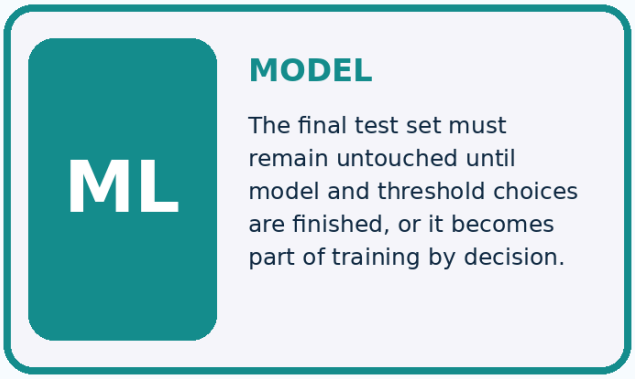
▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Identify contaminated evaluations and repair the workflow.

MODEL IDEA The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Team checks test accuracy after every parameter change. Those repeated choices adapt to the test set, contaminating it.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

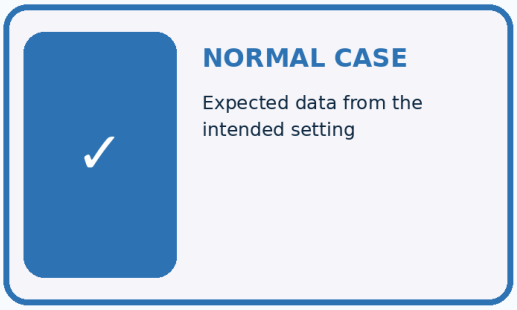
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

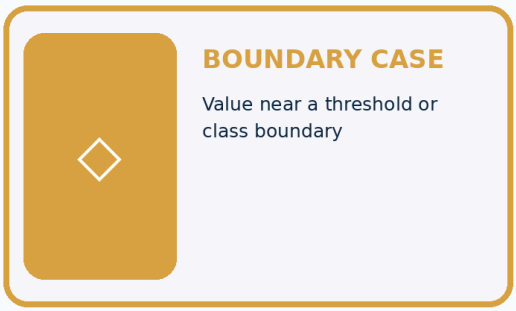
Compare a frozen final check with test-informed revision

> Renaming or reshuffling already inspected cases cannot undo their influence.

Four evaluation conditions



NORMAL Expected data and expected behavior.

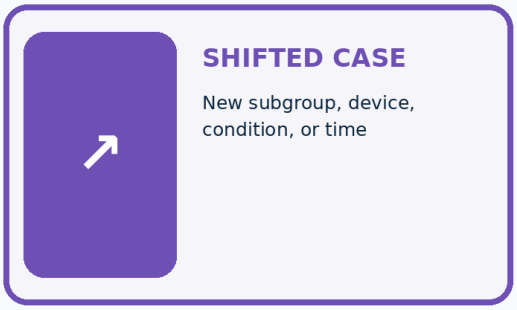


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's **claim**, and why?

Predict the supplied changed case

- Change: The team changes a threshold after inspecting the false negatives.

▼ **Check after predicting**

Prediction: This set is now development **evidence** for the revised model.

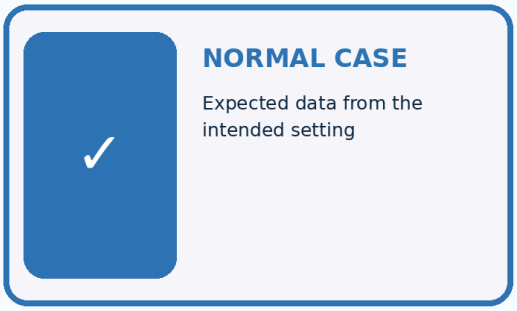
Reason: Using final outcomes to change the model breaks the untouched hold-out role.

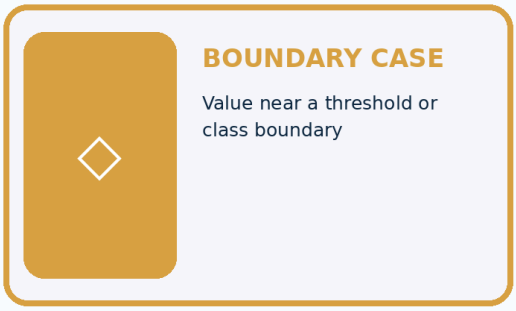
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

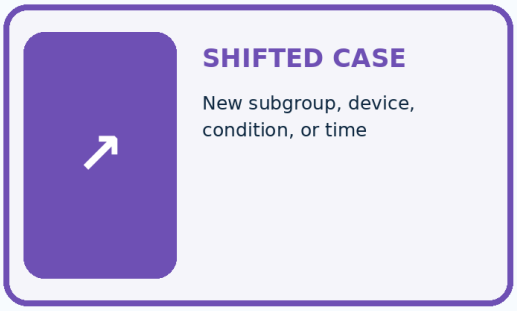
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the version boundary beside the metric calculation

> The same data can retain a historical result while losing untouched status for a revision.

Complete pictured counts and separate-version scope

Open the final hold-out only after recording the frozen model choice.

Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

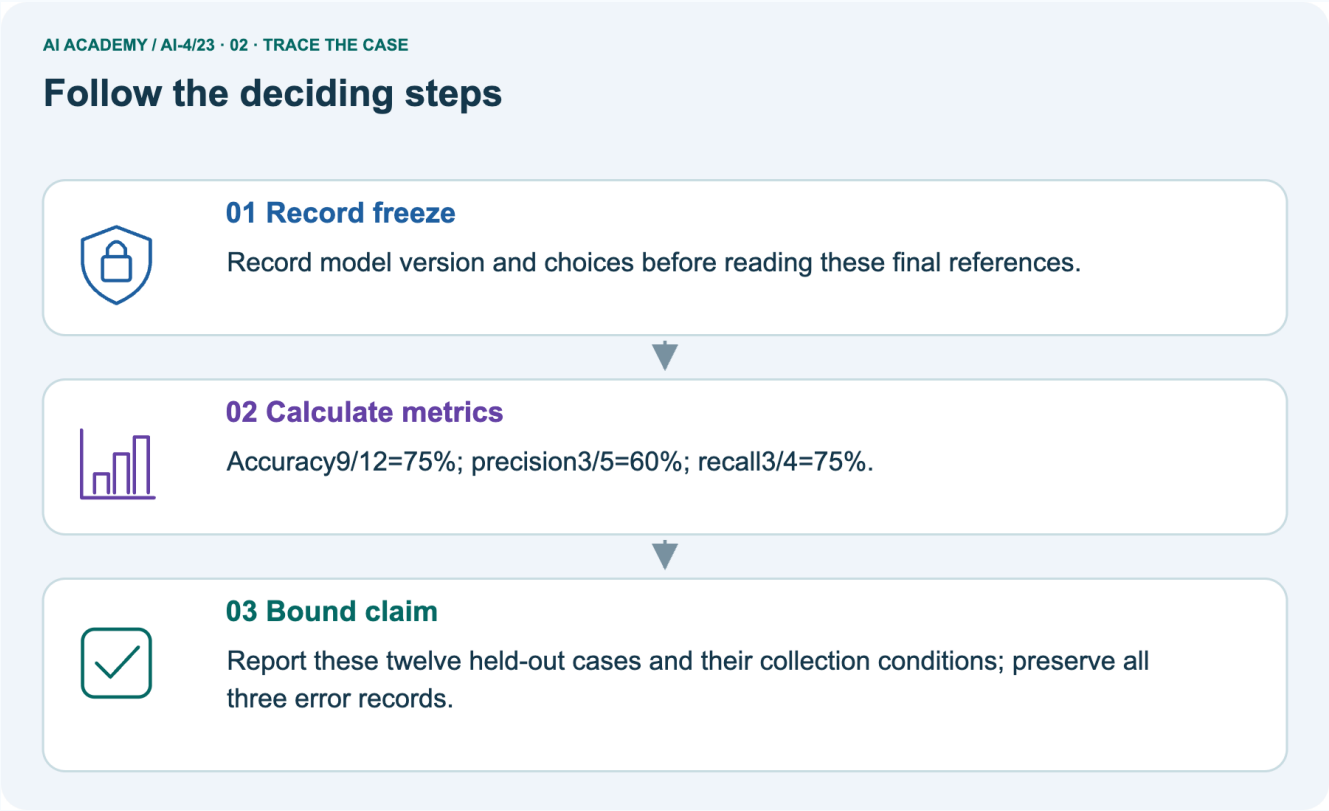
All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

Main20-case result is TN12,FP2,FN3,TP3:accuracy15/20=75%,precision3/5=60%,recall3/6=50%;always-clear baseline14/20=70%. Separate pictured12-leaf result is

TP3,FN1,FP2,TN6:accuracy9/12=75%,precision3/5=60%,recall3/4=75%. Equal accuracy/precision do not make the cohorts or recall identical. If the pictured false negative guides a new threshold,the set is development evidence for that revised version.

The historical frozen-version result remains recorded with its original scope; no revised-version test result is invented.

Trace freeze,metrics and claim limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

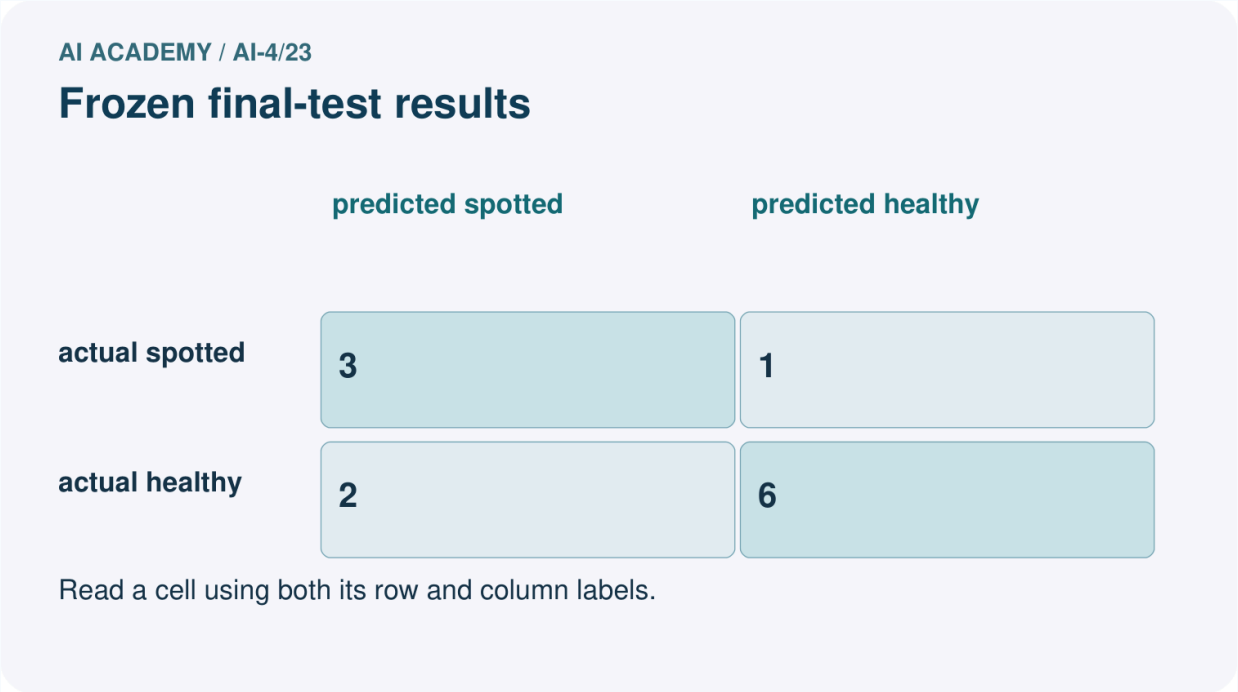
STEP	WHAT HAPPENS AND WHY
1. Record freeze	Record model version and choices before reading these final references.
2. Calculate metrics	Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.
3. Bound claim	Report these twelve held-out cases and their collection conditions; preserve all three error records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original hold-out workflow picture

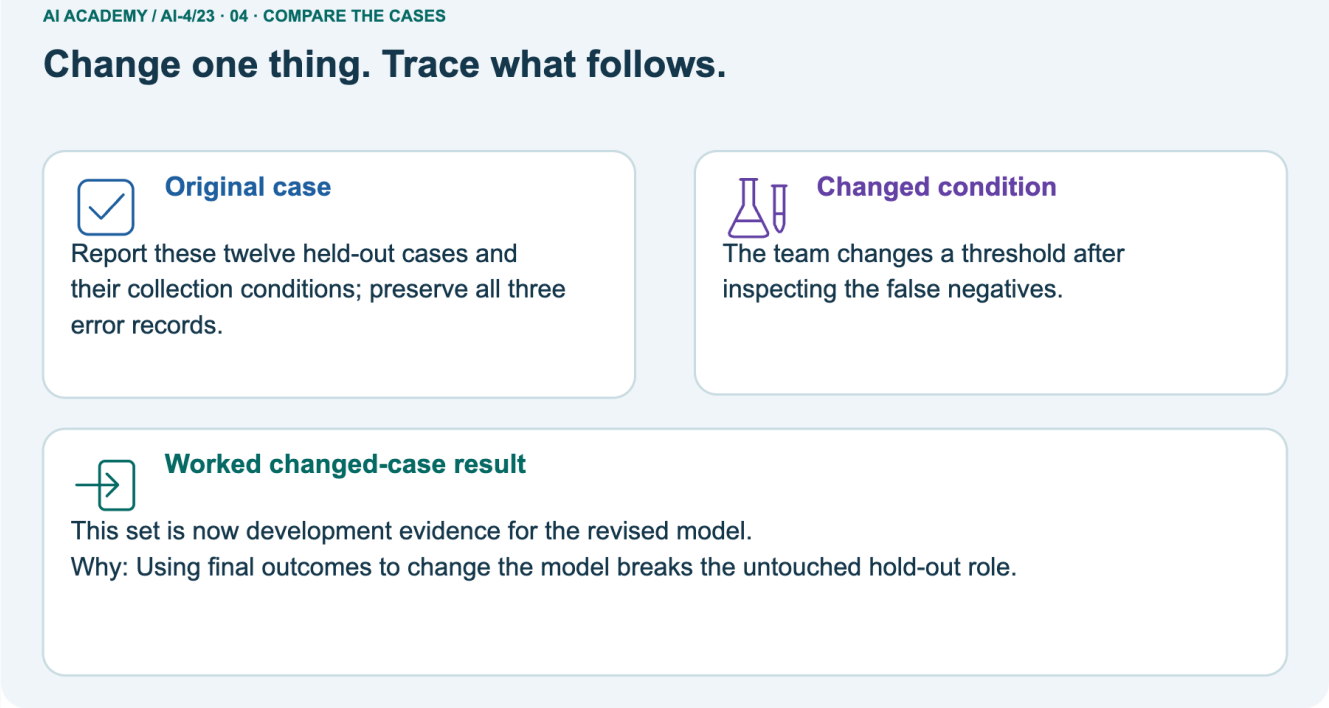


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict a threshold change after inspecting errors

- Change one thing: The team changes a threshold after inspecting the false negatives. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain new-evidence and sampling limits

- A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

- Read the labelled picture: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main baseline comparison question

Final testing and honest claims at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before labels	Freeze version and rules	Record test membership
During evaluation	Keep every planned case	Count failures explicitly
After evaluation	Publish full results	Restrict the claim

The test comparison must use the same cases and consistent rules. Report the denominator, reference-label process and sampling limitations. If the software fails on one case, report that failure; do not silently drop the case and keep the original headline denominator.

Why does 75 percent versus 70 percent not settle deployment? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 23 • WORKED EVIDENCE

1. Team checks test accuracy after every parameter change.
2. Those repeated choices adapt to the test set, contaminating it.
3. Use training for fitting and validation for model or threshold choices.
4. After freezing choices, open the holdout once and report the final estimate honestly.

STEP 1 Team checks test accuracy after every parameter change.

STEP 2 Those repeated choices adapt to the test set, contaminating it.

STEP 3 Use **training** for **fitting** and validation for model or threshold choices.

STEP 4 After freezing choices, open the **holdout** once and report the **final estimate** honestly.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

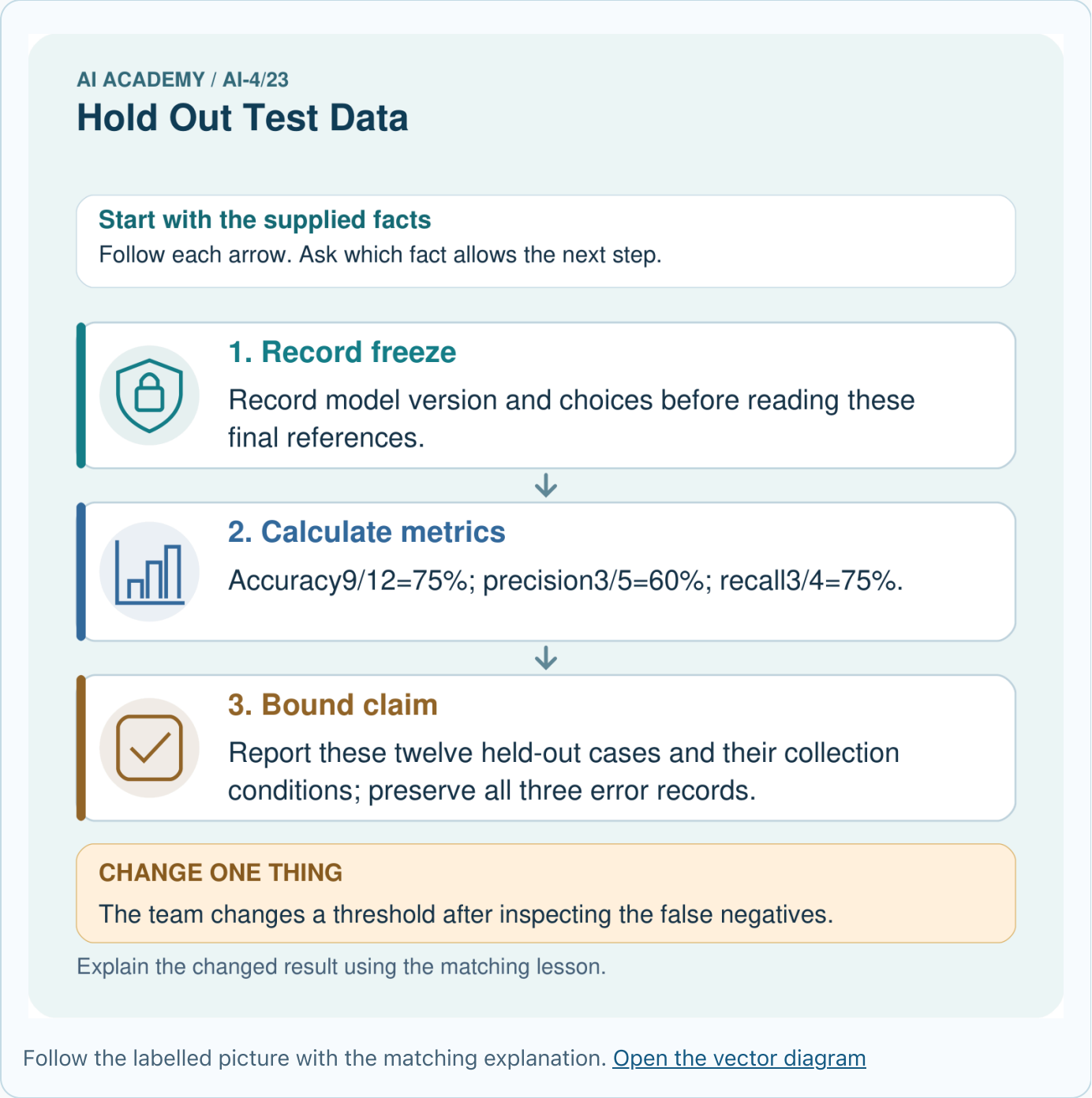
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hold Out Test Data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a test folder name mistaken for independence

> Record what information guided which choice.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Test data remains unbiased as long as the model never directly fits its labels.

Explain the limitation and next step

- A report can honestly say 15 of 20 correct on this fictional held-out set. It cannot say the model is 75 percent reliable for every child, school or future **input**. Do not fabricate more examples to make a result sound decisive.

> **One planned case returns no prediction. What must the report retain?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Test data remains unbiased as long as the model never directly fits its labels. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise an honest post-test development record

> Preserve the first result; do not relabel the revised score as independent.

Try supported W2

Use workbook W2 for the connected example “Final testing and honest **claim**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Trace the threshold revision and exposed-set role

After seeing three missed blocked cases, the team changes its threshold. The old test has now informed development. Preserve the first result and label later scores as post-test development.

A new independent test is needed for a new untouched-test **claim**; repeatedly splitting the same inspected rows does not erase what was learned.

Explain the evidence behind the claim

- Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain what the revised model still needs

- What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the fresh independent12-case count task

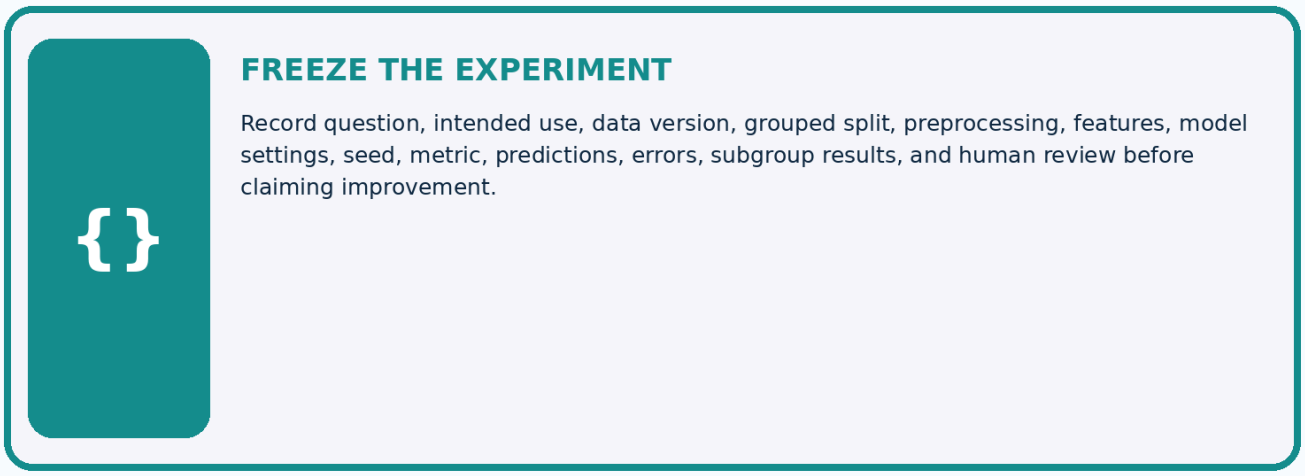
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Compare a baseline

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete split-integrity and final-report exercise

> Every workflow action,version record and evaluation pair needed for the court is supplied.

Complete frozen20-case record

Final-record case ID	Reference	Frozen V1prediction	Frozen always-clear baseline
F01	clear	clear	clear
F02	clear	clear	clear
F03	clear	clear	clear
F04	clear	clear	clear
F05	clear	clear	clear
F06	clear	clear	clear
F07	clear	clear	clear
F08	clear	clear	clear
F09	clear	clear	clear
F10	clear	clear	clear

Final-record case ID	Reference	Frozen V1prediction	Frozen always-clear baseline
F11	clear	clear	clear
F12	clear	clear	clear
F13	clear	blocked	clear
F14	clear	blocked	clear
F15	blocked	clear	clear
F16	blocked	clear	clear
F17	blocked	clear	clear
F18	blocked	blocked	clear
F19	blocked	blocked	clear
F20	blocked	blocked	clear

Freeze,baseline and no-fresh-data report

Freeze/exposure field · Complete fictional record

> **Freeze/exposure field:** Before-label record
>
> **Complete fictional record:** V1and its training-fitted preparation;blocked positive;clear/blocked axes;membershipF01–F20;accuracy,precision,recall;count every planned prediction and disclose failures.

> **Freeze/exposure field:** Baseline
>
> **Complete fictional record:** Previously fitted always-clear constant;do not choose it from F01–F20references. Original training counts are not supplied in this lesson.

> **Freeze/exposure field:** Observed result

Complete fictional record:
TN12,FP2,FN3,TP3:accuracy75%,precision60%,recall50%;baseline70%accuracy.

Freeze/exposure field: After-label revision

Complete fictional record: V2threshold changed using F15–F17misses. Preserve V1report;any V2score on F01–F20is post-test development. No V2predictions or improved score are supplied.

Freeze/exposure field: No fresh data available

Complete fictional record: Report the historical V1result and V2development status separately;state that no independent final estimate for V2is available. Do not manufacture a new score by resplitting inspected cases.

Eight complete workflow actions to judge

Workflow action to judge
C1 Freeze model,training-fitted transform,threshold,metrics,case membership and failure policy before label inspection
C2 Fit preprocessing on the combined training and final-evaluation features
C3 Run the frozen version and publish all counts,errors and limits
C4 Change threshold after reading missed positives,then report the same-set score as a revised final test
C5 Shuffle the already inspected final rows and call the new subset untouched
C6 Drop a missing prediction silently after seeing the score effect
C7 Compare V2and baseline on different row sets without saying so
C8 With no fresh cases,report V1historical result and V2post-test development without a final-performance claim

▼ Check reasoned judgments after trying

Complete workflow action	Judgment and evidence reason
C1 Freeze model,training-fitted transform,threshold,metrics,case membership and failure policy before label inspection	Consistent with the intended final-check workflow;still audit source quality and group separation.
C2 Fit preprocessing on the combined training and final-evaluation features	Evaluation information changes fitted state;split/group first and fit preprocessing only on training.
C3 Run the frozen version and publish all counts,errors and limits	Allowed reporting under the declared protocol;no deployment guarantee.
C4 Change threshold after reading missed positives,then report the same-set score as a revised final test	Development reuse;preserve original result and disclose that a fresh revised-version final estimate is absent.
C5 Shuffle the already inspected final rows and call the new subset untouched	Invalid independence claim;information exposure persists.
C6 Drop a missing prediction silently after seeing the score effect	Invalid denominator change;apply the declared failure policy and report exclusions/failures.
C7 Compare V2and baseline on different row sets without saying so	Not a matched comparison;record source versions,membership and policy differences.
C8 With no fresh cases,report V1historical result and V2post-test development without a final-performance claim	Honest bounded report;do not invent independent evidence.

Normal,identity-boundary,missing and revised-version probes

Complete final-test probe	Expected bounded check
Normal frozen F01–F20record	15/20correct,3/5precision,3/6recall;retain all5errors.
Boundary every case has a unique filename but related source groups cross sets	Unique names do not establish source independence;revisit the evaluation claim and grouping.

Complete final-test probe	Expected bounded check
Missing prediction on planned extra F21	Do not silently drop it;apply the predeclared policy and report attempted/valid/failure counts with the denominator. No F21reference is supplied.
Changed version selected after reading F01–F20outcomes	The inspected set is development evidence for V2;no independent revised-model score is provided.

Judge,repair,count and report

1. Split-Integrity Court: judge all eight complete workflow actions,point to the information crossing the boundary,and repair the evidence record. Verify the20-case table counts before comparing V1with the fixed baseline. Draft a report naming version,data,metrics and missed blocked cases. Then write the separate no-fresh-data statement for V2without inventing predictions.

   Public lesson fixtures illustrate these roles;actual learner-model final checks require genuinely unexposed,source-appropriate cases. Keep the selected new count task for an uncoached first attempt.

Retain evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Split-Integrity Court: classify workflow actions as allowed or contaminating, then redesign the evidence trail.

1. Write the hypothesis and expected **evidence** before running.

2. Split-Integrity Court: classify workflow actions as allowed or contaminating, then redesign the **evidence** trail.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: report metrics and a bounded claim for fresh counts

> Keep the frozen-version premise and each denominator explicit.

Complete independent counts and scope

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

Use the selected TN7,FP1,FN2,TP2for12new fictional cases,with blocked positive and clear/blocked reference-row/prediction-column axes. These differ from both the20-case main fixture and the pictured12-leaf counts. State the specific frozen-version scope and missing source-condition detail rather than claiming dependable performance in every setting.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the final-check evidence trail with the project

> A revised model should not inherit an earlier version’s untouched-test claim.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

Keep a project freeze record linking model/preprocessing versions,features,threshold,metrics,source groups,exact case membership and missing-prediction policy. Record who opened reference labels and which later changes used them. Preserve original results and label post-test revisions honestly.

A classroom procedure does not create clinical,agricultural or other consequential approval;the report’s claim must match the actual evidence and collection conditions.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

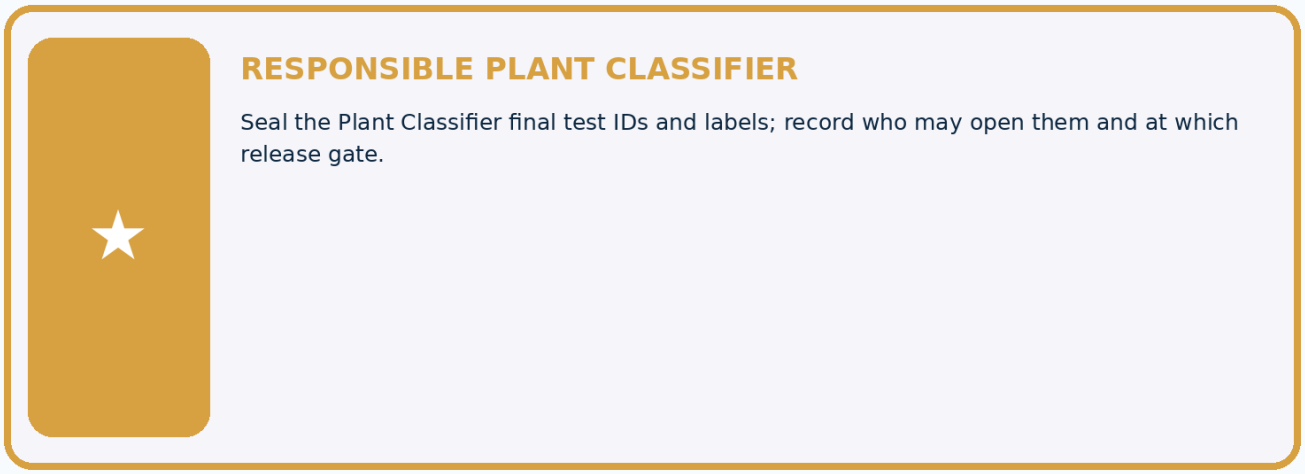
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish by naming the version,the exposure and the claim

> Explain the honest report when a revision has no fresh final evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: **Freeze** the full evaluation procedure before testing

> **My deciding step, evidence and one remaining question**
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
holdout	reserved unseen evaluation data	
contamination	test information influences choices	
model selection	choose among candidate models	
final estimate	performance measured once after choices freeze	

Closer-explanation terms

Word	Meaning
Freeze	Record a fixed model and procedure before testing
Held out	Kept separate from development decisions
Provenance	A record of where data and results came from

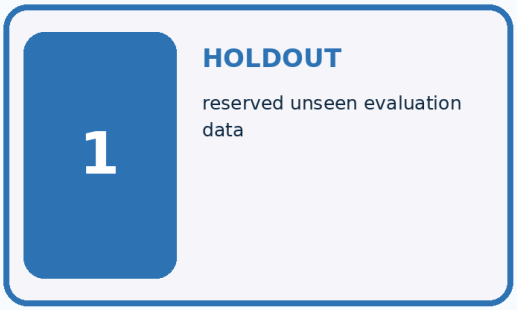
Word	Meaning
Claim	A statement whose scope must match its evidence

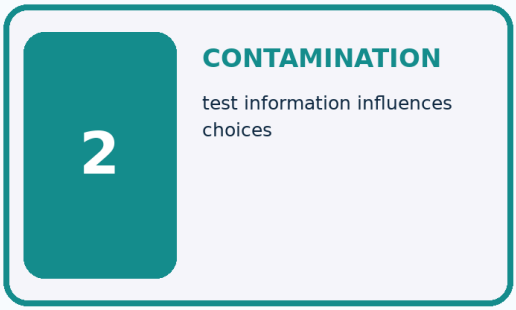
▼ Earlier wording, alternatives and picture transcripts

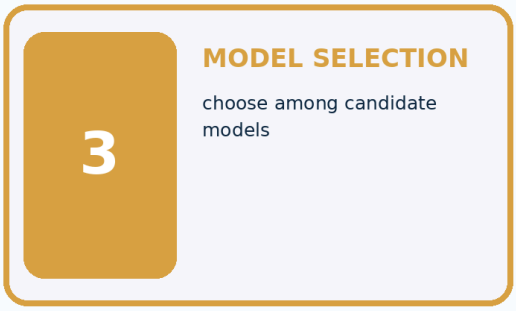
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Test data remains unbiased as long as the model never directly fits its labels.

Repair the **claim** using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **holdout**, **contamination**, **model selection**, **final estimate**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Test data remains unbiased as long as the model never directly fits its labels.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

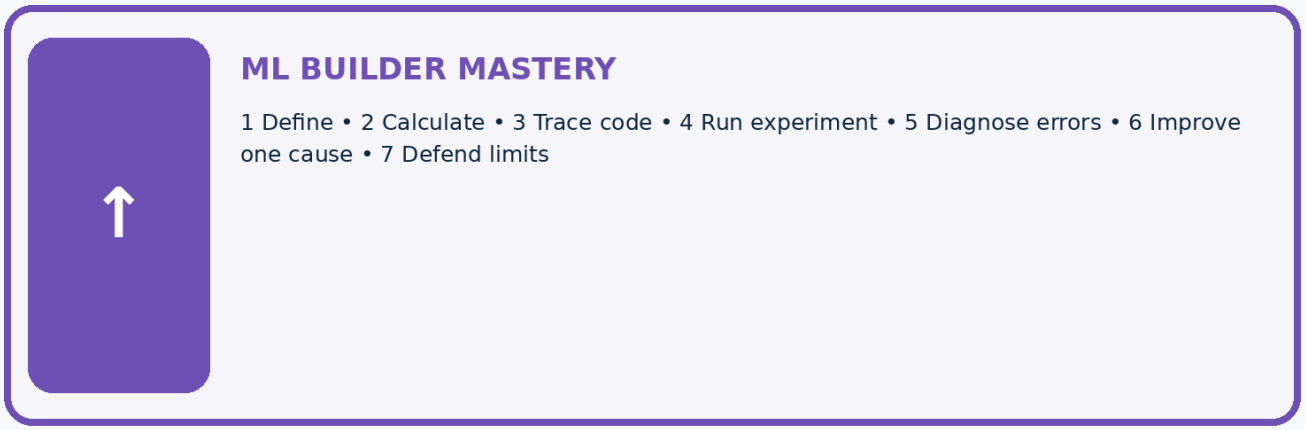
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh TN7/FP1/FN2/TP2metric-and-claim task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

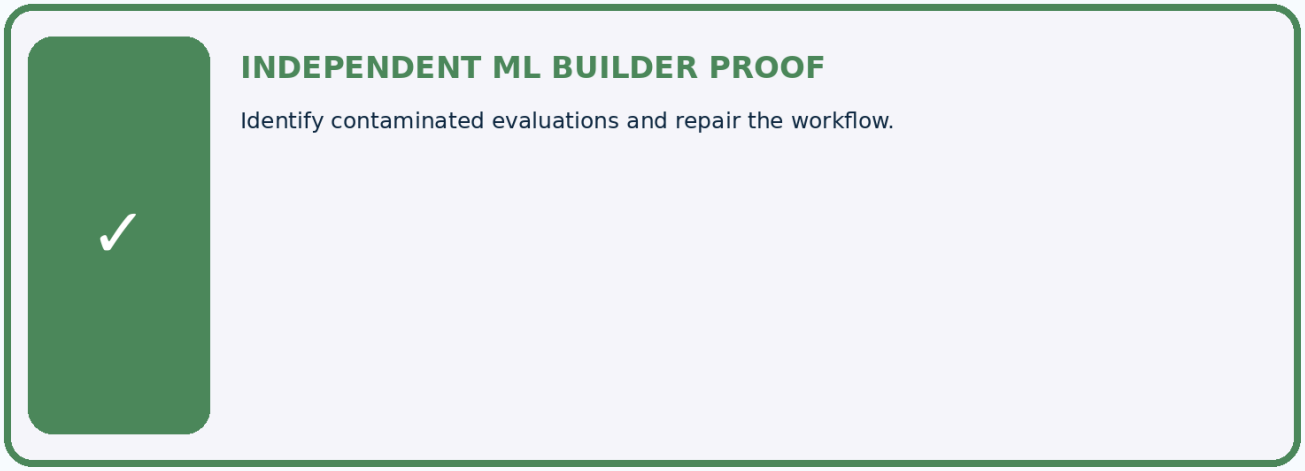
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Identify contaminated evaluations and repair the workflow.

NEXT BRIDGE Carry the experiment **evidence** into: Train, Validation, Test.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

Evidence or stage	Value or result
Reserved folder	Twenty leaves
Weekly inspection	Labels influence model choice
Actual role	Validation
Repair	Reserve genuinely unused final cases

▼ Earlier wording, alternatives and picture transcripts

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A repeatedly inspected test set is no longer final

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Freeze the full evaluation procedure before testing	Explain the deciding step, show a trace or test, and state one limit.
Report a held-out result with counts and limitations	Explain the deciding step, show a trace or test, and state one limit.
Recognize when test feedback has become development	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Open the final hold-out only after recording the frozen model choice.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Open the final hold-out only after recording the frozen model choice.

- Which labelled step matches this detail? Report these twelve held-out cases and their collection conditions; preserve all three error records.
- 1. Record **freeze**
- 2. Bound **claim**
- 3. Calculate metrics
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bound **claim** Report these twelve held-out cases and their collection conditions; preserve all three error records. Read the picture in order.

Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The team changes a threshold after inspecting the false negatives. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This set is now development **evidence** for the revised model.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: This set is now development **evidence** for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Repair and keep the record

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

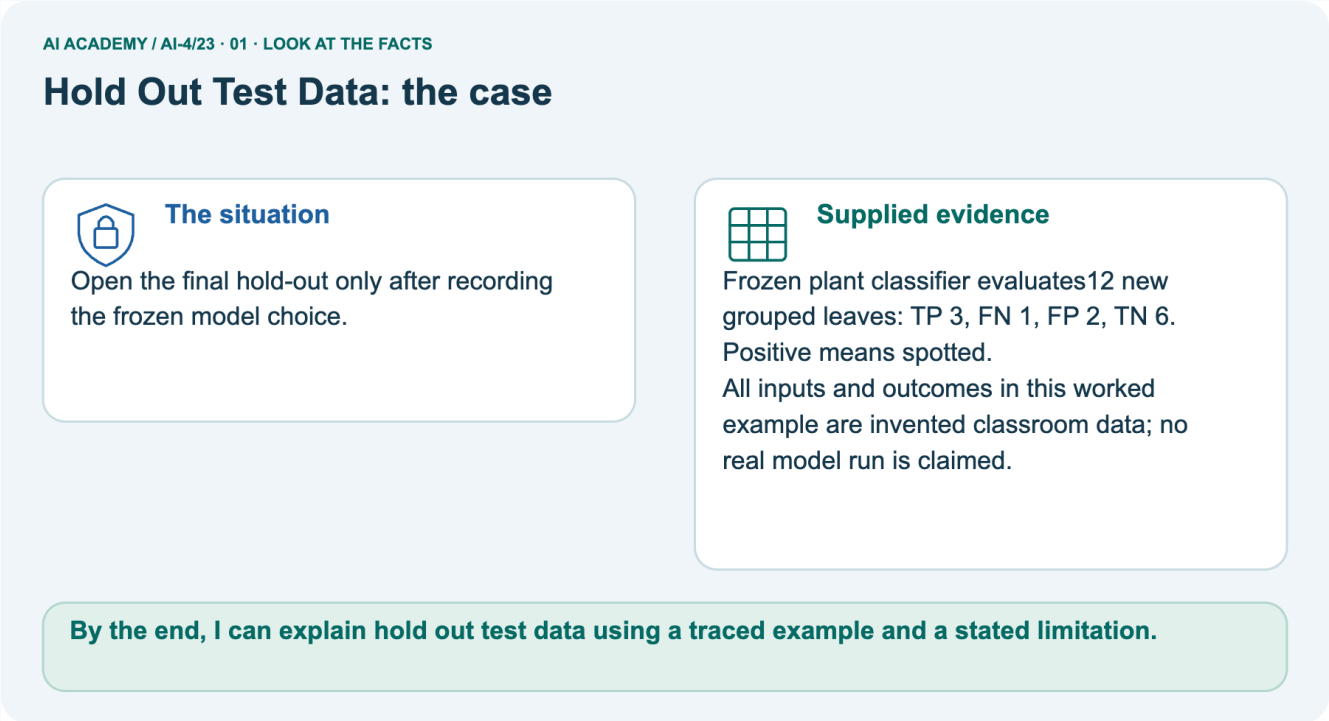
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Hold Out Test Data: the case



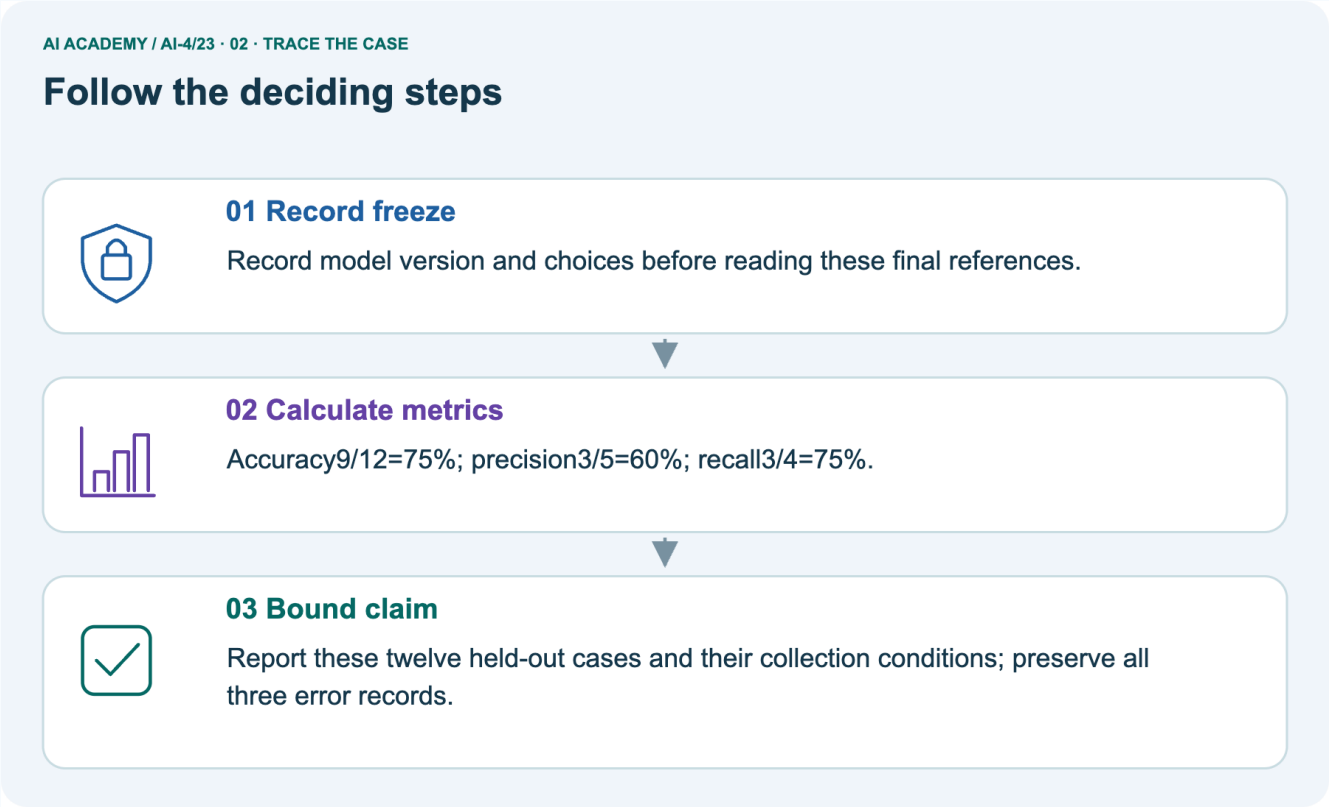
Follow the labelled picture with the matching explanation. Open the vector diagram

Open the final hold-out only after recording the frozen model choice. Supplied facts: Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

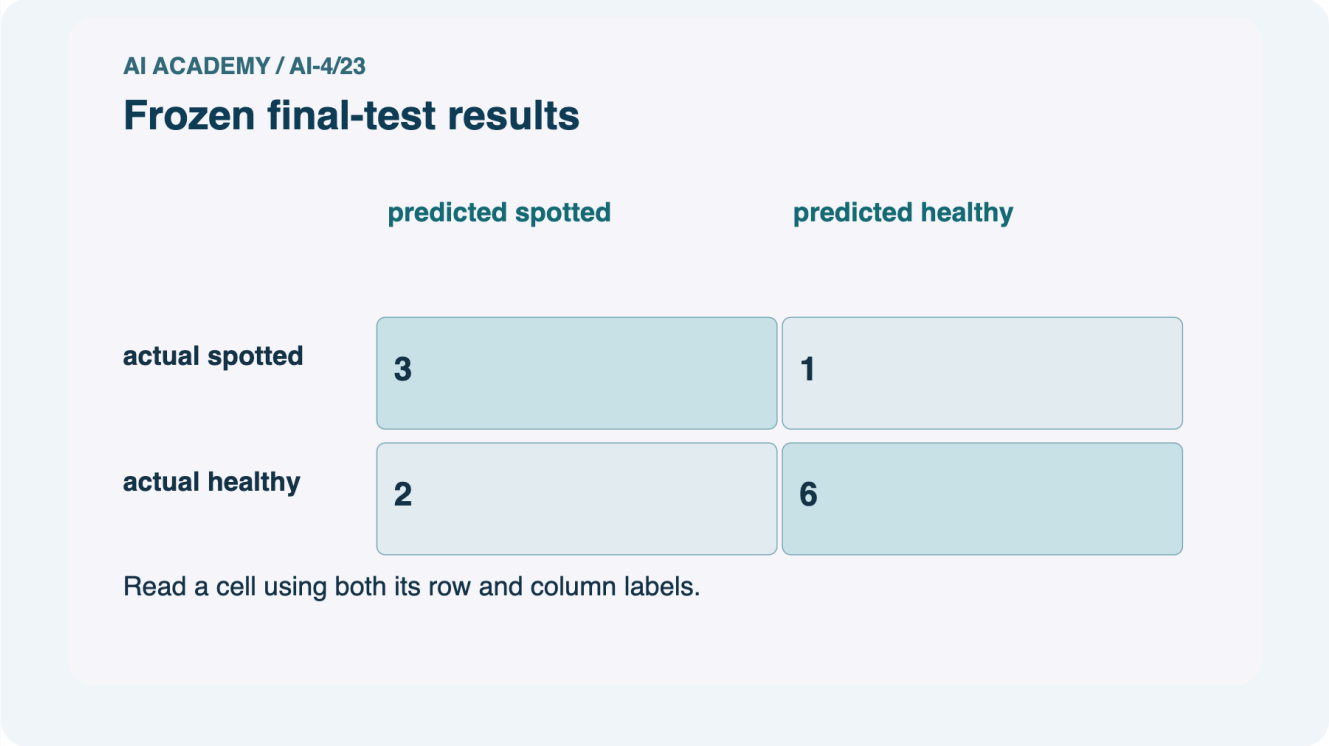


Follow the labelled picture with the matching explanation. Open the vector diagram

Record **freeze**: Record model version and choices before reading these final references. → Calculate metrics: Accuracy9/12=75%; precision3/5=**60%**; recall3/4=75%. → Bound **claim**: Report these twelve held-out cases and their collection conditions; preserve all three error records.

Point to each numbered stage and explain which recorded **input** it uses.

03. Frozen final-test results

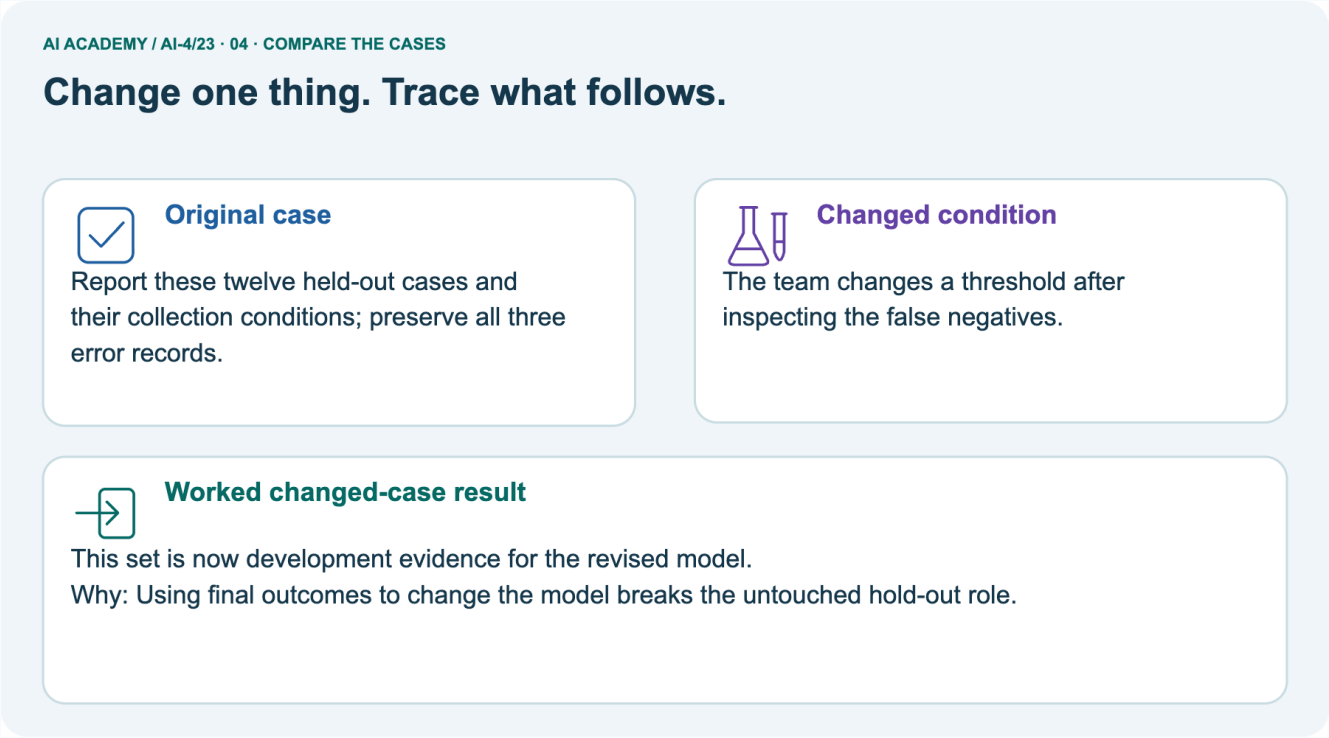


Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen final-test results. actual spotted: predicted spotted = 3, predicted healthy = 1. actual healthy: predicted spotted = 2, predicted healthy = 6.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



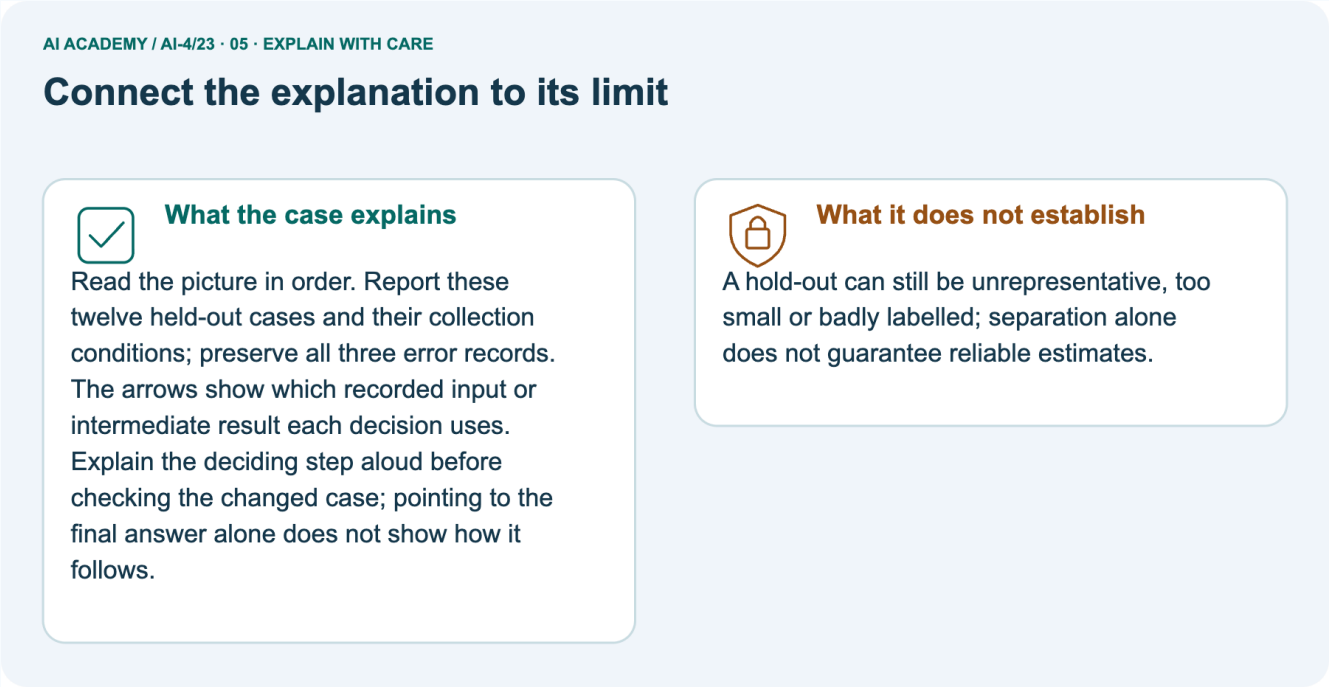
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report these twelve held-out cases and their collection conditions; preserve all three error records. Changed condition: The team changes a threshold after inspecting the false negatives.

Expected result: This set is now development **evidence** for the revised model. Reason: Using final outcomes to change the model breaks the untouched hold-out role.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

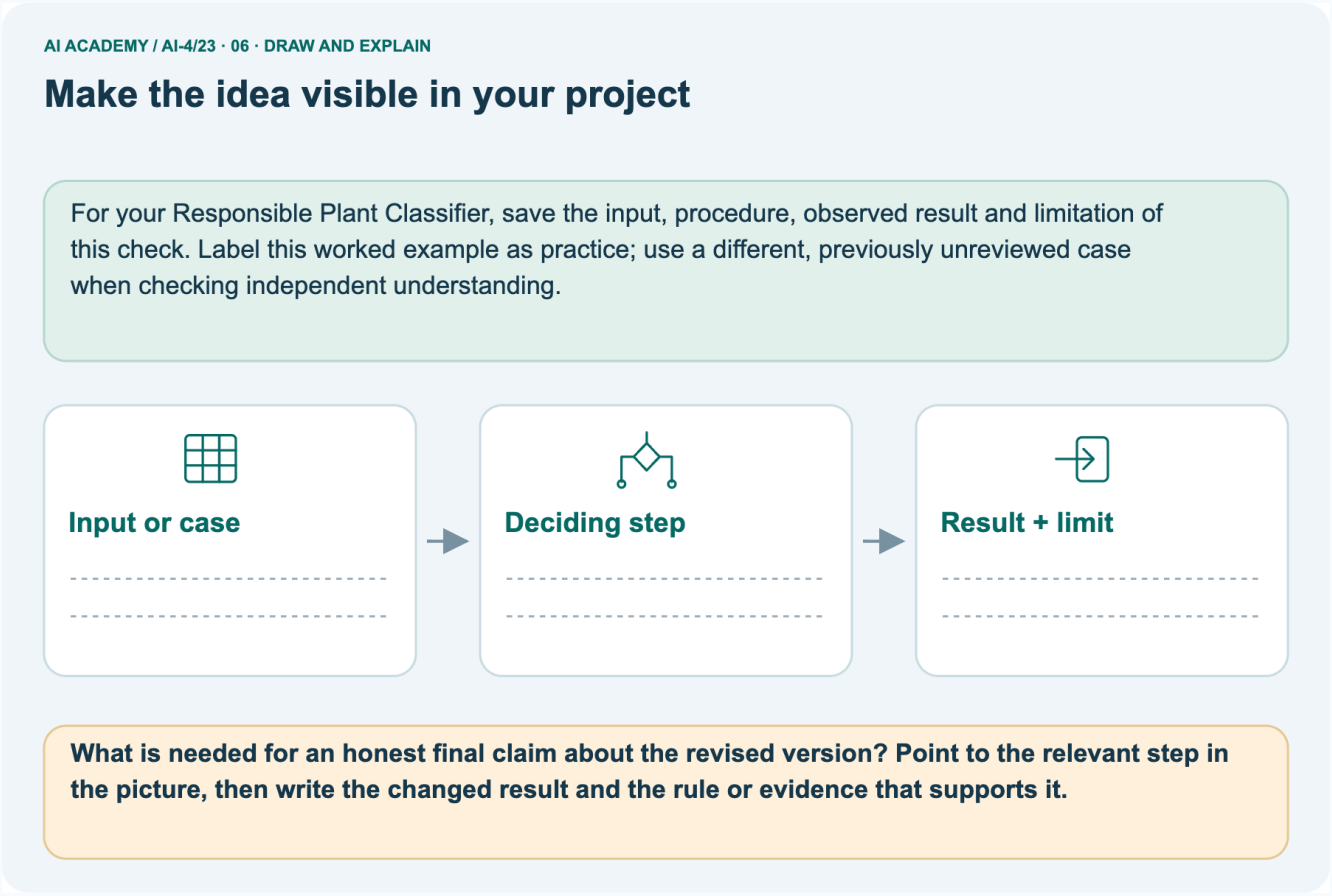
Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Hold Out Test Data: the case

02. Follow the deciding steps

03. Frozen final-test results

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.